

Decompose a Goal

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can distinguish a function definition from a function call.

I can match positional arguments to parameters in order.

I can reuse one function and test that it uses the supplied values.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 2: Decompose a Goal

Complex tasks become manageable when divided into testable subproblems.



THINKER MISSION

Break a school-morning helper into inputs, decisions, and actions.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

A script sets capacity = 12 and used = 4. What does print(capacity - used) display?

This week's end goal

By the end, I can decompose a calculation into a function with explicit arguments and a returned result.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use decompose a goal to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional supply cupboard records capacity and the number of occupied spaces. The remaining-space calculation is capacity minus used. Instead of copying the same instructions for every cupboard, define one function and call it with different values. A function makes a procedure reusable; its name alone does not guarantee that its instructions are correct.

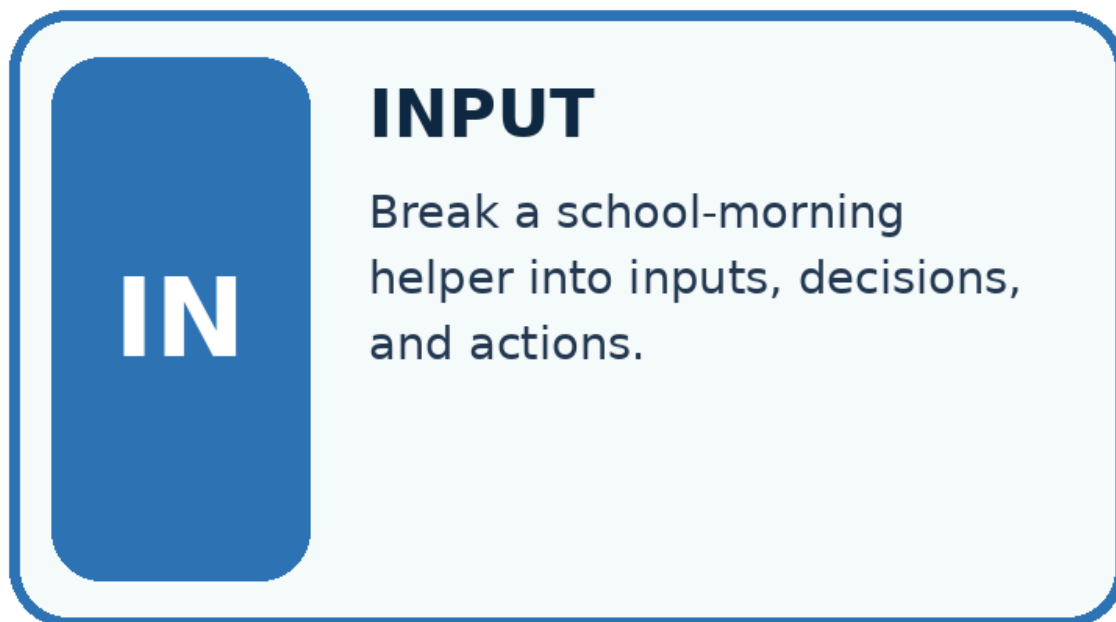
Our input contract requires nonnegative integer counts with used no greater than capacity. The example assumes these conditions; it does not validate them. This week the function prints its result. Week 9 will show how to return a value for later calculations.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Complex tasks become manageable when divided into testable subproblems.



OUT

OUTPUT

Break a school-morning helper into inputs, decisions, and actions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def show_remaining(capacity, used):  
    remaining = capacity - used  
    print(remaining)  
  
show_remaining(12, 4)  
  
show_remaining(10, 7)
```

Arguments matched in order

Call · Parameter values · Printed output

Call: show_remaining(12, 4)

Parameter values: capacity 12; used 4

Printed output: 8

Call: show_remaining(10, 7)

Parameter values: capacity 10; used 7

Printed output: 3

Each call runs the same body with the values supplied for that call.

def introduces the definition. The function name is followed by parameter names in parentheses and a colon. Four-space indentation marks the body. Executing the definition creates the function but does not run its body. The two unindented calls then run it, producing 8 and 3 on separate lines. A blank line helps readability; indentation determines the block.

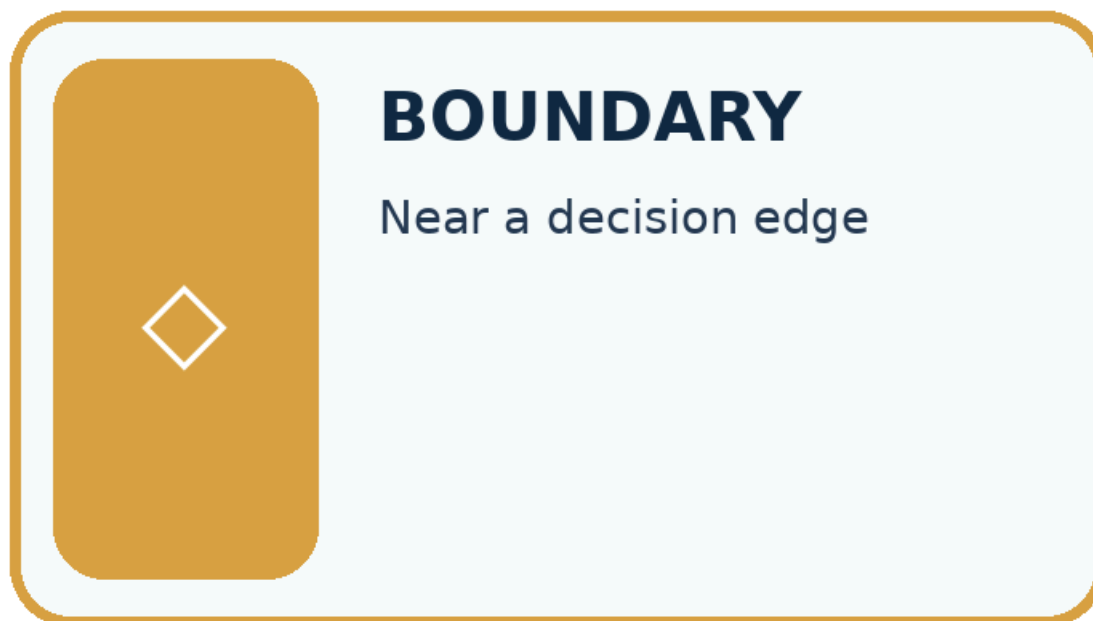
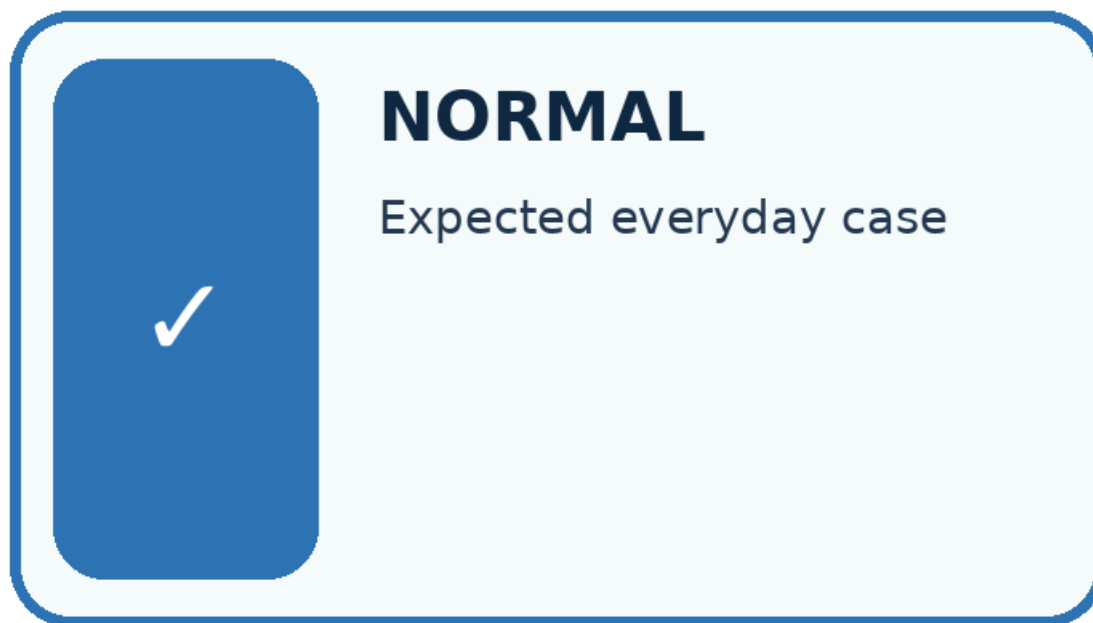
Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

A system is not understood until it survives more than one comfortable example.



NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Call labels(4,0) instead.

Prediction: The returned result is 0, and the caller prints 0.

Reason: Four shelves times zero labels per shelf is zero.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

DECOMPOSE

split a large task into
smaller parts

2

SUBPROBLEM

one smaller solvable task

3

DEPENDENCY

a part that needs another
part first

4

INTERFACE

how two parts exchange information

Word · My own example · Not this / boundary

Word: decompose

My own example: Break a goal into smaller parts with clear responsibilities and connections.

Not this / boundary: _____

Word: subproblem

My own example: A smaller part of a larger problem with its own inputs and intended result.

Not this / boundary: _____

Word: dependency

My own example: Something a step needs from another step before it can proceed correctly.

Not this / boundary: _____

Word: interface

My own example: An agreed boundary specifying how two parts exchange information or actions.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Function

Meaning: A named block of reusable instructions that runs when called

Word: Definition

Meaning: Code that creates a function and specifies its body

Word: Call

Meaning: An instruction that runs a function with supplied arguments

Word: Parameter

Meaning: A name in a function definition that receives an argument

Word: Argument

Meaning: A value supplied when calling a function

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 2 • WORKED EVIDENCE

- 1 Goal: help a learner leave home ready.
- 2 Parts: check time; check weather; check bag; choose reminder.
- 3 Dependency: weather must be known before the clothing suggestion.
- 4 Test each part separately, then test the whole flow.

1. Goal: help a learner leave home ready.
2. Parts: check time; check weather; check bag; choose reminder.
3. Dependency: weather must be known before the clothing suggestion.
4. Test each part separately, then test the whole flow.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Execution event · What happens

Execution event: Execute the definition

What happens: Create the function; no output yet

Execution event: Call with 12 and 4

What happens: Match parameters; subtract; print 8

Execution event: Call with 10 and 7

What happens: Match new values; subtract; print 3

Execution event: After the second call

What happens: Continue with the next statement

In `show_remaining(12, 4)`, 12 and 4 are arguments; capacity and used are parameters. Positional arguments match from left to right. `show_remaining(4, 12)` produces -8, showing why order matters; it also violates our written input contract. Use a fresh file with the definition before its calls.

Predict the output of `show_remaining(9, 9)` followed by `show_remaining(9, 0)`. Name the parameter receiving each argument. What would be printed if both calls were removed?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decompose a Goal

A fictional library computes total spine labels needed for several equal shelves.

The facts we will use

Define `labels(shelves,per_shelf)`: return `shelves*per_shelf`.

Call `needed=labels(4,6)`, then `print(needed)`. No other side effects are defined.

Decompose a Goal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Call with arguments

The call supplies shelves=4 and per_shelf=6 locally.



2. Function body

Multiply 4 x 6 to calculate 24.



3. Return value

return sends 24 back to the caller; it does not print.



4. Caller prints

needed receives 24; print(needed) displays 24.

CHANGE ONE THING

Call labels(4,0) instead.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Call with arguments

What happens and why: The call supplies shelves=4 and per_shelf=6 locally.

Step: 2. Function body

What happens and why: Multiply 4x6 to calculate 24.

Step: 3. Return value

What happens and why: return sends 24 back to the caller; it does not print.

Step: 4. Caller prints

What happens and why: needed receives 24; print(needed) displays 24.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A function separates one named job from the surrounding program. Arguments supply data for this call; return supplies its result. A print statement in the caller is a different action. Clear parameter names help detect swapping or misusing inputs without assuming every function prints automatically.

What this example does not establish

This function assumes compatible numeric inputs; it does not reject negative shelves or verify real shelf contents.

Read the labelled picture: What happens visibly if the caller stores needed but omits print(needed)?

Change one thing: Call labels(4,0) instead. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Decomposition means writing every tiny action before understanding the goal. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

For each call, record the supplied arguments, expected printed output and observed output. Include

different values that produce different answers, plus a valid zero-result case. Call order controls output order; it does not change the subtraction rule. These functions use supplied data and written instructions, not a pattern learned from examples.

A learner tests only `show_remaining(12, 4)` and says the function works for every cupboard. Give a second valid call that exposes the fixed-number bug, state its expected output and explain what passing two tests still cannot prove.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
def show_remaining(capacity, used):  
    remaining = 12 - 4  
    print(remaining)  
  
show_remaining(12, 4)  
  
show_remaining(10, 7)
```

This faulty function prints 8 twice. Its fixed numbers ignore the arguments. Replace 12 - 4 with capacity - used, then rerun both calls. Testing only the first call would miss the bug because its expected answer happens to be 8.

For this two-parameter definition, `show_remaining(12)` raises `TypeError` because an argument is missing. Correct the call to supply both values. Naming a function without calling it does not execute the body: the parentheses in `show_remaining(12, 4)` are part of the call.

Practise with less help: Follow the execution order

Use workbook W2 for the connected example “Functions and arguments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Functions and arguments. Proposed explanation: “Writing `def` runs the body immediately”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What happens visibly if the caller stores needed but omits print(needed)?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Cut a familiar task into 4–6 cards. Order dependencies, name each input and output, and test whether replacing one card breaks the whole system.

Cut a familiar task into 4–6 cards. Order dependencies, name each input and output, and test whether replacing one card breaks the whole system.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Separate reader profile, book data, score calculation, ranking, and explanation.

THIS WEEK'S CHECKPOINT Separate reader profile, book data, score calculation, ranking, and explanation.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Give the recommender separate validation, eligibility and scoring functions whose inputs and results can be tested independently.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Break a school-morning helper into inputs, decisions, and actions.

CHALLENGE Break a school-morning helper into inputs, decisions, and actions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A class wants to recommend one of five library books.

The question

Decompose the goal into checkable steps.

Worked explanation

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Decompose a recommendation into records

Evidence or stage · Value or result

Evidence or stage: Preferences

Value or result: Collect reader choices

Evidence or stage: Traits

Value or result: Describe five books

Evidence or stage: Scoring

Value or result: Apply the declared rule

Evidence or stage: Decision

Value or result: Handle ties and explain

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

14. Try a new case independently

Independent case: Create a new function independently

Each sealed box contains 6 counters. Define `show_stock(boxes, loose)` to print the total counters using its two parameters. Assume nonnegative integer arguments. Call it with (3, 2), (0, 2) and (3, 0). Predict each output and explain why `show_stock(2, 3)` is a different case.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Separate reader profile, book data, score calculation, ranking, and explanation.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional library computes total spine labels needed for several equal shelves.

By the end, I can · How I will show it

By the end, I can: Distinguish a function definition from a function call

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Match positional arguments to parameters in order

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Reuse one function and test that it uses the supplied values

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Distinguish a function definition from a function call

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.